LLL-T-431b

OCTOBER 22, 1959

SUPERSEDING
Int. Fed. Spec. LLL-T-00431a(COM-NBS)
August 7, 1958 and
Fed. Spec. LLL-T-431
August 20, 1937

FEDERAL SPECIFICATION

TILE, CORK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Types and classes.—Cork tile furnished under this specification shall be of the following types and classes, as specified in the invitation for bids:

Type I.—Unfinished.

Type II.—Finished.

Class 1.—Wax, lacquer, or resin finish.

Class 2.—Clear plastic film finish.

1.2 Size.—The size of the tile shall be as specified in the invitation for bids. See 6.2.

1.3 Thickness.—The thickness shall be 1/2, 5/16, 1/4, 3/16, 1/8, or 3/32 inch as specified in the invitation for bids.

1.4 Edge.—The edges of the tile shall be square or beveled as specified in the invitation for bids.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-636—Boxes, fiber.

PPP-T-97 — Tape; pressure-sensitive, adhesive, filament reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available *without charge* at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Bid sample.—A sample of each type, class, and color shall be submitted with bids to show color, finish, and surface characteristics.

3.2 Material.—Cork tile shall be made from a good quality of clean cork substantially free from hardback and foreign material. The particles of cork shall be thoroughly and uniformly bonded together by means of a suitable baking process. The tile may contain a thermosetting binder.

3.3 Color, finish, and surface characteristics.—The wearing surface shall be smooth and clean. The tile may be furnished in varying and variegated shades of natural cork. The color, finish, and wearing surface characteristics shall match the sample mutually agreed upon by buyer and seller.

3.3.1 *Type I.* — Type I tile shall have a smoothly sanded wearing surface without added finishing materials.

3.3.2 *Type II.*

3.3.2.1 *Class 1.*—The wearing surface of class 1 tile shall be factory treated with a wax and/or lacquer and/or resin.

3.3.2.2 *Class 2.*—The wearing surface of class 2 tile shall be factory finished with a clear plastic film. When tested as described in 4.3.8, the film thickness shall average not less than 0.003 inch.

3.3.3 The back of both types I and II tiles may be treated with a water-repellent coating which will not adversely affect the adhesion of the tile to the floor.

3.4 Weight.—The tile shall weigh not less than 26 pounds per cubic foot.

3.5 Initial and residual indentation.—When tested as described in 4.3.5, the initial indentation and the residual indentation shall not exceed the values given in table I:

TABLE I.—*Indentation*

Thickness	Initial indentation, maximum	Residual indentation, maximum
Inch	*Inch*	*Inch*
3/16 and over	0.020	0.007
1/8 or less	0.013	0.004

3.6 Disintegration.—The tile shall show no evidence of disintegration when subjected to the boiling water test in 4.3.7.

3.7 Squareness.—The tile shall not deviate from squareness by more than 0.025 inch when tested as described in 4.3.6.

3.8 Tolerances, dimensions.

3.8.1 *Length and width.*—A tolerance of ± 0.25 percent shall be permitted in the length and width.

3.8.2 *Thickness.*—A tolerance in the thickness of ± 0.005 inch in square-edged tile and ± 0.010 inch in the beveled-edge tile shall be permitted.

3.9 Workmanship.—The tile shall be free from defects which may affect its appearance or serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—Unless otherwise specified in the invitation for bids, all tiles of the same type, class, size, thickness, and kind of edge presented at one time shall be considered a lot for the purpose of inspection and tests.

4.1.2 *Sampling for inspection.* — Unless otherwise specified in the invitation for bids, a random sample of tiles shall be taken for inspection from each lot of tiles in accordance with table II. A unit shall be one tile.

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TABLE II.—*Sampling for inspection*

Number of units in lot	Number of Units in sample	Acceptance number (defectives)	Rejection number (defectives)
300 or under	3	0	1
301 to 800	5	0	1
801 to 1300	9	1	2
1301 to 3200	15	1	2
3201 to 8000	25	2	3
8001 and over	35	3	4

4.1.3 *Sampling for test.*—Unless otherwise specified in the invitation for bids, a random sample of tiles shall be taken for test from each lot of tiles in accordance with table III. A test unit shall consist of three tiles. The sample for test shall be taken from the sample for inspection in accordance with a random process.

TABLE III.—*Sampling for test*

Number of *units in lot*	Number of test units in sample	Acceptance number (*nonconforming*)	Rejection number (*nonconforming*)
5000 or less	3	0	1
5001 to 25000	7	1	2
25001 and over	15	2	3

4.2 Inspection.

4.2.1 *Nonmilitary agencies.*—Each unit in the sample for inspection, 4.1.2, shall be inspected for type, class, edge, color, surface characteristics, workmanship, packing, and marking.

4.2.2 *Military agencies.*—Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.3 Tests.

4.3.1 *Conditioning.*—Tests may be made on tile conditioned under prevailing atmospheric conditions except in cases of dispute. In case of dispute, tests shall be made on tile which has been conditioned not less than 46 hours in an atmosphere of 50 ± 4 percent relative humidity and at a temperature of 23° ± 1.1° C. (73.5° ± 2° F.). The physical tests shall be conducted in the same atmospheric conditions.

4.3.2 *Thickness.*—Each tile in the test unit shall be calipered at three equally spaced places at least ¾ inch from the cut edges with a dial gage micrometer or equivalent. The dial gage indicator shall have a flat presser foot not more than 0.75 inch or less than 0.20 inch in diameter. The other flat bearing surface (anvil) shall be no smaller than the presser foot. The foot shall exert a pressure of 20 ± 5 pounds per square inch in the measuring range. The gage shall be graduated in 0.001 inch. Each of the nine measurements shall fall within the limits given in 1.3 and 3.8.

4.3.3 *Linear dimensions.*—A scale or tape graduated to 0.01 inch or less shall be applied over the tile in such a manner that the tile is not distorted. The length or width shall be read from the scale to the nearest 0.01 inch and the value recorded. Each linear dimension of each tile in the test unit shall be measured at two equally spaced places. Each of the twelve measurements shall be within the limits given in 1.2 and 3.8.

4.3.4 *Weight.*—The thickness and linear dimension of one tile from each test unit shall be measured and the weight determined on a suitable balance to the nearest 0.1 gram. The weight per cubic foot of the tile in the test unit shall be calculated from the volume and weight of the specimen.

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4.3.5 *Indentation.*

4.3.5.1 *Specimen.*—The specimen shall consist of a portion of the test unit at least 2 by 2 inches if rectangular, or 2 inches in diameter if circular in shape. The thickness of the specimen shall be the thickness of the material undergoing test.

4.3.5.2 *Apparatus.*—The apparatus shall be as follows:

4.3.5.2.1 An apparatus consisting of an indenter acting under a total weight of 100 pounds, a dial indicator for measuring the depth of indentation, a rigid-metal plate for supporting the specimen, and a rigid-metal frame for supporting the weight and indenter.

(*a*) The indenter consists of a steel bar rigidly supported vertically in such a manner that the face of the lower end (foot) contacts the specimen perpendicularly to the face of the specimen when the specimen is placed in a horizontal position on the rigid metal plate. The face of the indenter foot that contacts the specimen is flat and buffed smooth but not rounded. The foot of the indenter shall be 1.125 inch in diameter. The upper end of the indenter is provided with a weight-releasing device for applying the load without impact to the indenter and for activating a dial indicator from which the depth of penetration of the indenter can be read. The weight of the indenter shall not exceed one pound.

(*b*) The dial indicator attached to the upper end of the indenter is equipped with a scale graduated to read the depth of penetration to 0.001 inch.

(*c*) The metal plate for supporting the specimen shall be rigidly fixed in a horizontal position in the framework of the apparatus and shall have a smooth, flat surface.

4.3.5.2.2 A stopwatch or other timing device that will indicate the time in seconds.

4.3.5.3 *Procedure.*

4.3.5.3.1 The thickness of the specimen at the center shall be determined as described in 4.3.2 and the value recorded to the nearest 0.001 inch as T_1.

4.3.5.3.2 *Initial indentation.* — The specimen shall be placed flat on the supporting plate of the apparatus with the wearing surface upward. The indenter foot shall be lowered gently, without impact, until it contacts the surface of the specimen at the point where the thickness measurement was made. The scale on the dial indicator shall be set at zero, and within 5 seconds the total load applied to the specimen by means of the weight release. The total load shall be maintained for 600 ± 10 seconds. At the end of the required time of the application of the load, the indentation shall be read from the scale of the dial indicator and the value recorded as the initial indentation of the specimen.

4.3.5.3.3 *Residual indentation.*—At the end of the 10-minute period the load shall be immediately and completely removed. Sixty minutes ± one minute after the removal of the load the thickness shall again be measured at the indented spot as described in 4.3.2 and the value recorded to the nearest 0.001 inch as T_2. (The residual indentation is equal to T_1 minus T_2.)

4.3.5.4 One specimen from each of the three tiles in the test unit shall be tested. The initial and the residual indentation of the test unit shall be the average of the values obtained from the specimens tested.

4.3.6 *Squareness.*

4.3.6.1 *Specimen.*—The specimen shall be a single tile from the test unit.

4.3.6.2 *Apparatus.* — The apparatus consists essentially of a dial gage and two index strips mounted in a flat bedplate so as to form an angle of exactly 90 degrees, and a

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reference gage for setting the dial indicator at zero.

4.3.6.2.1 The dial gage is located in the lower left-hand corner of the apparatus with the stem extending horizontally so that the foot will contact the edge of a tile held in contact with the two reference strips. The dial is graduated to read to 0.001 inch and has a stem travel of approximately one inch. The contact foot of the indicator stem is 0.25 ± 0.01 inch in diameter which exerts a total force on the tile of not more than 3.0 ounces.

4.3.6.2.2 A reference gage for use in setting the dial indicator to zero and checking the angle of the index strips.

4.3.6.3 *Procedure.*—The tile shall be placed on the bedplate with the entire length of one of the edges held flush against the horizontal index strip and approximately one inch from the vertical left-hand index strip. The stem of the dial indicator shall be held in the retracted position and the tile moved to the left while maintaining contact with the horizontal index strip, until the left-hand edge of the tile is in contact with the vertical index strip. While pressing the tile firmly against the vertical index strip, without distortion of the material, the dial indicator stem shall be released gently, without impact, until the foot contacts the edge of the tile. The out-of-square shall be read from the indicator scale to the nearest 0.001 inch. Measurements shall be made on all four corners of the tile and the maximum value recorded to the nearest 0.001 inch.

4.3.6.4 The three tiles in each test unit shall be tested. All of the values obtained on the tiles tested shall be within the limits specified in 3.7.

4.3.7 *Resistance to boiling water.* — One specimen not less than one square inch in area from each test unit shall be floated on the surface of boiling water for 3.0 ± 0.1 hour. At the end of the flotation period, the specimen shall be examined for evidence of disintegration.

4.3.8 *Thickness of plastic finish, type II, class 2.*

4.3.8.1 *Specimen.*—The specimen shall be a disk 1⅛ inch in diameter.

4.3.8.2 *Apparatus.*—The apparatus shall consist of a microscope equipped with devices capable of making measurements accurately to 0.0005 inch. A suitable apparatus is a microscope equipped with an appropriate eyepiece and objective to provide a magnification of approximately 45X, a mechanical stage, and a ruled glass micrometer disk.

4.3.8.3 *Procedure.*

4.3.8.3.1 *Preparation of specimen.*—Each specimen shall be cut approximately in half with a sharp single-edge razor blade or equivalent, with the wearing surface or plastic coated side up. The cutting edge of the blade is held perpendicular to and drawn along the 1⅛-inch axis of the disk. With a single stroke the plastic film is cut completely through its thickness and the break continued by snapping the disk between the fingers.

4.3.8.3.2 *Thickness measurement.*—Three specimens shall be taken at random from each of three tiles. Either half of each specimen shall be mounted, cut side up, on the mechanical stage of the microscope. The cut surface of the cross section shall be brought into focus and at the midpoint along the 1⅛-inch axis, the thickness of the plastic film shall be determined from the number of rulings on the glass micrometer disk (in the eyepiece) required to cover the film thickness. If the film thickness is such that it does not all lie in the field of view, the specimen shall be moved by means of the mechanical stage until the entire film thickness passes through the field of view and the micrometer disk rulings counted during the movement recorded.

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4.3.8.4 *Results.* — Nine specimens from each test unit shall be tested. The average thickness of the wearing surface of the test unit shall be the average of all the values obtained. The average thickness shall be recorded to the nearest 0.0005 inch.

4.4 Compliance criteria.

4.4.1 *Inspection.*

4.4.1.1 *Nonmilitary agencies.*—Any tile in the sample for inspection failing to conform to the requirements for which the inspection was made, 4.2.1, shall be considered defective. If the number of defective tiles in the sample exceeds the acceptance number for that sample specified in table II, the entire lot of tile from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.4.1.2 *Military agencies.*—Compliance of the lot with respect to inspection, with the exception of packing, packaging, and marking 4.5, shall be as described in 4.4.1.1.

4.4.2 *Tests.*—Any tile in the sample for tests failing to conform to the requirements for which tests were made, 4.3.1 to 4.3.8 inclusive, shall be considered defective. If the number of defective tiles in the sample exceeds the acceptance number for that sample specified in table III, the entire lot of tile from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.5 Military agencies, inspection of preparation for delivery. — The preservation, packaging, packing, and marking of the tile shall be examined and tested to determine conformance with the requirements of section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with Military Standard MIL–STD–105, with inspection level L–7 and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 **Civil agencies.**—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

5.1.1 *Packing.*

5.1.1.1 *Level C.*—Unless otherwise specified in the invitation for bids, tile shall be packed in crates or packages of such construction as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

5.1.2 *Marking.*—Unless otherwise specified in the invitation for bids, shipping containers shall be marked with the name of the material, the type, class, size, thickness, color, and the quantity contained therein as defined by the contract or order under which shipment is made, the name of the manufacturer, the name of the contractor, and the number of the contract or order.

5.2 Military agencies.

5.2.1 *Packing.*

5.2.1.1 *Level A.*—The tile shall be packed in close-fitting fiberboard boxes conforming to Federal Specification PPP–B–636 (class 3). Filament reinforced pressure sensitive adhesive tape conforming to Federal Specification PPP–T–97, applied in accordance with the appendix thereto, may be used in lieu of steel strapping to reinforce boxes. The gross weight of a box shall not exceed 70 pounds.

5.2.1.2 *Level B.*—The tile shall be packed in close-fitting fiberboard boxes conforming to Federal Specification PPP–B–636 (class 1). The gross weight of a box shall not exceed 90 pounds.

5.2.1.3 *Level C.*—The tile shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates.

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5.2.2 *Marking.*—Marking shall be in accordance with Military Standard MIL-STD-129. When special markings are required they shall be specified in the contract or order.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, size, thickness, and kind of edge, 1.1.
(*c*) Finish, color, and surface characteristics, 3.3.
(*d*) Level of packing required, 5.2.

6.2 Sizes.—Cork tile is commercially available in the following sizes, in inches:

Square: 6 by 6, 9 by 9, 12 by 12.
Rectangular: 6 by 12 and 12 by 24.

6.3 A 4-inch or 6-inch cork cove base is also available.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served for the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Transportation description.—Transportation description and minimum weight applicable to this commodity are:

Rail:
Tile, flooring, cork, felt base, asphalted, decorated, painted or plain.
Carload minimum weight 30,000 pounds.

Motor:
Tile, flooring, cork, felt base, asphalted, decorated, painted or plain.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO. 15419

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